

-
-
-
-
-

Chapter 7

Vendor Scheduling Overview



Introduction

Companies frequently need to purchase goods on a repetitive basis, and the accepted method of raising discrete purchase orders and receiving goods can be burdensome.

In addition, there is often a requirement to source supply from different vendors with whom a contract may or may not exist. The Vendor Scheduling module allows you to set up contracts and then source the periodic supply from one or more vendors, using MRP, MPS or DRP.

If any one item appears on more than one contract, the Vendor Splitting Rules will be defined. These rules show how suggested requirements are allocated across multiple vendors and / or across multiple contracts for the same vendor.

Only items that have the Vendor Scheduling indicator set on the Item Stockroom file are processed by Vendor Scheduling.

Features of Vendor Scheduling

The features of the Vendor Scheduling Module are:

- Full purchase contract control
- Optional contract restrictions (e.g. maximum value)
- Support for automated multi sourcing with splitting rules
- On-line review and maintenance of schedules
- Schedule send via fax or EDI.
- Supports re-ordering of consumable items (including automatic option)
- Fully integrated with other modules

Contract Maintenance

A contract may be maintained for purchase of one or more items from a vendor with details of effective dates, currency, total quantity, and value. More than one contract may be entered into for the same vendor and items. Items may be attached to the contract with appropriate price structures, sourcing rules and contract maximums for quality and value. Appropriate standard enquiries may be called to assist in this task.

Contract Control

When a buyer has identified a source of supply and has completed all contractual negotiations, a vendor contract will be established to formalise the agreement. Each contract in Vendor Scheduling is for one vendor, although there may be many contracts for each vendor. The contract is divided into two main components;

- The contract header details which relate to the contract as a whole
- The contract item details which are specific to an individual item

Individual companies can authorise different users to have access to Contract Maintenance and can further restrict those users who have the authority to authorise a contract. Once established and authorised the contract is printed and sent to the vendor for confirmation.

Each contract can optionally have further restrictions defined on the header and at item level. Traditionally many companies released blanket purchase orders that were open forever and had no quantity limits. The potential problem with this scenario is, if an employee ordered 5,000,000 items instead of 5,000 (due to a keying error) the liability is enormous. Using contract restrictions available within Vendor Scheduling can minimise these liabilities, particularly as the planning and re-ordering process becomes more automated which leads to reducing the non-value adding activities of an organisation.

The restrictions available are:

- Effective from and to dates
- Maximum values (in whichever currency)
- Maximum quantity

Each of the above can be applied to header and / or item detail level. As these restrictions are firm, i.e. the users cannot order more than the restriction allows, Vendor Scheduling includes a report to show which contracts are due to expire within X weeks, more than Y% complete and so on. The manual and automatic release options also produce exception reports showing contracts where dates or quantities have expired.

There is a lot more data that can be defined against a contract. For example, users can define up to three authorisation periods to override those set at company level (typically Fabrication, Material, and Forecast) as well as stipulating whether the schedule will be printed, and how often, transmitted by fax or EDI and whether history will be kept. The Vendor Scheduling module will also automatically track and update cumulative receipts, cumulative quantity ordered, and cumulative quantity returned for each item and each contract.

Multi Sourcing Rules

Many organisations have a policy of ordering certain high volume items from more than one vendor. This is supported in Vendor Scheduling. For each item a sourcing table may be set up which will be used to calculate vendor and quantity for any supply requirements. The sourcing rules, at the highest level, determine the effective dates of the sourcing rules and whether to source by price, vendor percentage or overall percentages.

At item level the rules determine the proportions that are placed on each vendor or more particularly against each contract and any batching rules that apply. The individual vendor's rules may or may not be attached to a contract.

The choice of three algorithms to be used to split the requirements for an item across multiple vendors are:

- Least cost
- Percentage of individual requirements
- Percentage of overall requirements

Least Cost

This rule allocates the particular requirement to the supplier which has the lowest price for the quantity required. There are two places this information is held, either for a particular contract, or from the item / supplier profile. The process is as follows:

For the requirement, find the lowest price for the particular quantity, firstly looking at the contract price breaks, then the item / supplier profile price breaks. If only one supplier has the lowest price, then allocate the entire requirement to this supplier. The prices are converted to base currency and compared at base currency level before deciding the vendor to chose.

If two suppliers have the same price for the same quantity and one of them is the preferred supplier on the item details (in inventory), then this supplier will be used for the requirement. If both are defined, then the primary one will be used first, then the secondary etc. If neither supplier is defined then the system will use the supplier with the highest year to date value (from the P/L period balances files), else the system will allocate in alphabetical order.

Percentage of Individual Requirements

This rule allows the allocation of one individual requirement to more than one vendor.

The multi-sourcing details file defines the criteria which will determine the allocation of the requirement. A vendors supply values will only be included in the overall picture in circumstances where a contract is entered if the effective dates on the contract (or contract item) make the item current. Each criteria will be considered in turn.

The apportionment value indicates the proportion of the requirement to be allocated to the specified vendor, if no other restrictions to the supply quantities are applied. This is the first criteria considered in the processing to determine the initial amount to be allocated to each vendor. All the proportion values for the vendors are to be accumulated (where a contract is entered, the value is only to be included if the contract item is current, which is determined by checking the dates on the contract and contract items file). Each individual vendors value is then divided by this total and the result multiplied by overall item requirement to give a basic requirement per vendor. This value will not be rounded until the final figure is derived. If proportion values are not entered, then progress down the list of vendors in sequence attempting to allocate all to the first vendor if possible.

Each vendor will be considered in turn for a complete set of rules, before progressing to the next vendor in the sequence.

After determining the apportioned value, the contract cumulative quantities to date must be checked (from the contract item details file), if the item is purchased against a contract, to ensure that the contract maximum has not been exceeded. If the contract quantity has been exceeded, the system will allocate to this supplier up to the contract limit, and attempt to allocate the remainder to other suppliers. If this is not possible, the system will retrieve the flag from the company profile which indicates if contract quantities can be exceeded, if this is 1, then the outstanding amount will be allocated to the original supplier, else the supply cannot be satisfied. The contract values can only be exceeded once, so if on performing this last potential reallocation, the contract 'ordered to date' values are already greater than the 'maximum quantity' of the contract item, then no further requirements can be added to the contract.

If a contract item maximum quantity is entered, this value must not be exceeded. If the derived value above exceeds this amount, then the maximum is to be ordered and the remainder allocated to the next vendor in the sequence, if this exceeds his maximum, the same is to apply to all vendors on the list. If all requirement has not yet been satisfied, then the remainder is to be added to the preferred vendor defined at the multi-sourcing rule header.

Minimum Quantity

This is the minimum quantity the vendor will accept on any individual requirement. If the amount is allocated to this vendor does not reach this amount, then the system will ignore this supplier and move to the next in the sequence if the round to minimum indicator is not set.

Round to Minimum

If this flag is set to 1, and the quantity allocated does not reach the minimum quantity, then the system will round up the quantity to the minimum amount. E.g. if the amount to be allocated is 300, and the minimum quantity is 5000, the system will allocate 5000 to this supplier.

Round to Multiple

If this field is entered, then the system will round the amount allocated up to a whole number multiple of this amount.

If no rounding has occurred to the amount originally allocated, the system will round the quantity up to the next multiple of the purchase unit of measure.

After allocating an amount to the first vendor, the system reapportions the amount of the original requirement left to the remaining vendors and allocates as with vendor 1. This is repeated down the list in sequential order until the original requirement is satisfied. If the end of the list is reached and there is still some demand unsatisfied, then this is added to the preferred vendor defined on the multi-sourcing rule header file.



Note: If rounding (or reallocating any 'left over'), causes the maximum quantity to be exceeded then there is no recalculation, or any attempt at correction for this particular circumstance.

Percentage of Overall Requirements

This is very similar to the proportion of individual requirements, but the amount is allocated in accordance with the overall allocation amount to date for the particular rule. These values are maintained at multi-sourcing rule item level by schedule release.

This rule allows each individual requirement to be allocated to a particular vendor as a whole, but the vendor may vary.

The system retrieves the quantities to date for each item as recorded on the multi-sourcing rule detail file. It calculates the total for all vendors and divides each vendors quantity by the total to determine the proportion allocated to each vendor to date. Next the system will calculate the required proportion in the same manner as for the individual requirements (dividing the individual proportions on the multi-sourcing rules by the total for all proportions). Next, the actual to-date proportions are subtracted from the multi-sourcing rule proportions and the vendor with the greatest difference is determined. The entire requirement is allocated to this vendor.

The maximum rule does not apply.

Round to minimum / round to multiple apply as for rule 2 above.

If the allocated amount falls below the vendors minimum, and the flag to round up to minimum is 0, use the vendor who is second in the list of furthest away from the annual percentage, and so on. If the requirement falls below the minimum fall to all suppliers, and the rounding flags are 0, then the whole amount is allocated to the preferred vendor.

Create Schedules From MRP / MPS or DRP

With the base data established, the supplier scheduler reviews the results of the latest MRP / MPS / DRP planning processes and then runs the Vendor Scheduling function. The scheduler then reviews the results of the process with the Schedule Maintenance facility. This will show all schedules that have a source of supply problem that should be corrected before releasing the schedule.

Automatic Schedule Release and Schedule Maintenance

Auto Release

Vendor Scheduling offers the capability to fully automate the reordering process. Following an MRP planning run, an automatic update of supplier schedules can be initiated which will action all the MRP purchase suggestions for items controlled by Vendor Scheduling and purchase order lines are created. The automatic update function also provides a simulation facility which prepares a pre-list report of all schedules without actually confirming the process. The report lists contracts that have insufficient quantity remaining to fulfil the requirements of this planning cycle.

Processing

The schedules will be read in the sequence of the vendor / item / stockroom / contract. For each schedule only the current schedule will be processed (since all past schedules will have the history flag set to 1).

If the selection is set to only include schedules due for release then only process the schedule if the expected date of this schedule, on the schedule header is less than or equal to the assumed release date which has been entered.

Historic schedules that have not been issued or have not updated purchase order files will be purged in a separate function.

The processing of a schedule involves the update of the purchase orders. Once a schedule has been processed its status will be set to 'update / to issue' so that it is picked up by the print run. No updates will occur if the pre-list run type is taken.

Purchase Order Creation and Update

For each schedule, the date of the end of the FAB authority will be obtained from an ascending hierarchy of contract line and company profile. The schedule details will be read up to this date and converted into a purchase order.

If there is one purchase order for the contract then the purchase order will be obtained from the contract header. If there is one purchase order for all items in this run the purchase order will be the next number from P/O company profile. If there is one purchase order for each item in this run a new purchase order will be retrieved for each item.

Purchase orders will be created in the currency from a hierarchy of contract and vendor and in a unit of measure from a hierarchy of contract details and item purchase unit.

Landed costs will be attached if they exist at item / supplier level.

Schedule Maintenance

If schedules are generated that contain errors; e.g. insufficient quantity available on the contract, then these schedules will be reviewed using the Schedule Maintenance facility. All schedules available for release are able to be viewed and amended using this facility.

Schedules are printed and can be sent to suppliers using any combination of three different ways: post, by direct fax link or by EDI. The choice of which method is

defined on the contract for each vendor / item combination. The schedules are printed in a 'one item pre page' vertical format on A4 size layout. Vendor Scheduling also offers the capability to send schedule data directly by fax or EDI.

Method of Use

Supply suggestions are converted into Vendor Schedules by comparing the pricing and sourcing parameters, from one or more vendors, to determine the preferred vendor(s) at any one time. The pricing and sourcing data may be contracted or may be for items not under contract. Similarly the same item may be under two different contracts for the same vendor.

In addition to MRP, MPS or DRP, Inventory re-ordering will look at the sourcing rules for items that are multi-sourced and create a schedule albeit for a single delivery, when the stock levels reach the re-order point. Items not defined as vendor scheduled on the item stockroom file will not be processed by vendor sourcing.



Warning: It is critical that all MRP purchased items that are sourced through Vendor Schedules have MRP filters associated to not allow reschedule in, reschedule out, increase quantity, or decrease quantity. The only allowable values are for suggest delete. This will allow the Sourcing Rules and Contracts within Vendor Scheduling to action the demand requirements correctly.

The MRP runs, including net change, may be run any number of times between sourcing runs, since vendor sourcing will always pick up suggestions from the last run.

Once the source of supply has been determined you may review the schedules for all vendors / contracts and the individual vendor's schedule with the ability to override the system generated quantities and dates. Thus there will be no conflict between the number of times vendor sourcing is run and schedule review / release.



Warning: It is critical that the schedule print program is run after release and before the next vendor sourcing run.

After reviewing the schedule the user may release it, in which case purchase order lines will be created or amended and the new schedule will be ready to be sent to the customer.

Schedules are released and printed and sent to the supplier by post, fax or EDI.

Facilities are provided to automate the schedule sourcing, release and print / send functions so as confidence in the system grows the purchase officer(s) spends less time managing the system and more time on exceptions and other value adding activities.

The Vendor Sourcing Process

A batch run processes the source data from MRP, MPS, DRP, and inventory re-ordering.

Whenever run is selected, then the last run date and time for the user will be displayed.

MRP and MPS

This is a batch run which performs the multi-sourcing. The source data files for MRP and MPS are identical.

The function reads the supply file by company and active model code (retrieved from the MRP company profile field), retrieving the company, stockroom, item, planner, quantity, and due date / time (demand / supply date). Only those records which satisfy any selection criteria entered, have a record status not equal to processed, and are purchase items with the vendor schedules indicator set on the Item Stockroom profile will be selected for sourcing.

DRP

This functions in a similar manner to the MRP / MPS run, however, included in the selection will be the code or DRP centre entered on the selection screen.

The selections to be included are exactly the same as for the MRP / MPS items, but also select only those items within the entered distribution centre.

Inventory Reordering

The process to run Inventory ROP calculations to create vendor sourcing records as opposed to requisitions, is to allow items which are not controlled by MRP etc. to be purchased against a contract, or with multi-sourcing rules applied.

Any items which satisfy the criteria for inclusion into the schedule files are processed by vendor sourcing.

Item Stockroom

Only item records from the item stockroom file that have a vendor sourcing indicator set to 1 will be processed by vendor sourcing. This means that for some items in a stockroom their purchase is controlled by Requisitions and some items may be controlled by Vendor Scheduling.

Running Multi Sourcing

Each module source record is passed through the multi-sourcing algorithms to ensure the correct vendor is selected. Even if the item has no multi-sourcing algorithms the requirements are still to appear on the schedule file. This will allow the requirements to be reviewed by the schedule maintenance function.

If an item has a contract, but no multi-sourcing rules, then only the contract maximums are to be checked.

Output

The two output files are the schedule header and schedule detail files.

After the allocated amounts have been decided for each vendor, the schedule records are created, with the appropriate priority if required. The most important

priority has the highest number. If a particular schedule has more than one priority applicable, then the highest numbered priority is to apply.

The priority is user defined on the purchase management parameters file, type PRTY. The number to be updated on the schedule header file as the priority is the value field from the parameters file.

The following priorities are defined on the parameters file in descending order:

95=Invalid Vendor Code

90=Exceeded Contract / Not Ordered

80=Contract Unauthorised

70= Exceeded Contract / Ordered

60=Exceeded Contract / Ordered on Alternative Vendor

50=Significant Change Period 1 - for future use.

40=Significant Change Period 2 - for future use.

30=Significant Change Period 3 - for future use.

20=Significant Change Period 4 - for future use.

10=Significant Change Period 5 - for future use.

05=Supply placed on Preferred Vendor

00 (Blank)=Normal

Priorities 95, 90, 80, 70, and 60 have already been decided when running the multi-sourcing rules. Priority 05 is to apply if the multi-sourcing rules have 'failed' and supply have been placed on the preferred vendor, without any problems with contract cumulatives. The priority codes and associated descriptions are held on the Inventory Descriptions file type PRTY.

Enquiries

There are four main enquiries which will allow you to review the schedule information on the system.

Contract Enquiry

This allows you to view the contract details. The enquiry is solely to provide static information concerning the contracts.

Contract Item Enquiry

This enquiry allows you to view the Contract Item Details.

Sourcing Rules Enquiry

This enquiry allows you to enquire on multi sourcing rules set up for particular items.

Schedule Enquiry

This allows you to view details relating to a schedule that has been previously released and sent.

Reports

There are a number of reports which you can use to keep track of your schedules.

Contract Report

This allows you to list the contract details. The report is solely to provide static information concerning the contracts.

Contract Exception Report

This report allows you to determine those contracts which are about to expire, reach the limit etc. By careful selection of criteria those contracts which are reaching the end of their life, or are close to the maximum values defined can be easily identified.

Multi Sourcing Report

This report allows you to report on multi sourcing rules set up for particular stockroom and / or items.

Schedule Exception Report

This option allows you to list, by exception, the schedules which have been created. This will enable you to easily identify those items which may need manual intervention prior to the scheduling being issued.

Schedule Summary Report

This report allows you to list the created schedules in a summary format selected by particular criteria to limit the report.